Mounia Touil

SWE 6783 Assignment 3

1. Study Design Scenario:

This study is designed to show how a software quality assurance team experimented to investigate how the density of inline code comments affects the time a developer needs to fix a bug in that code. 30 developers participated in a between-subjects study, with 10 assigned to each condition. Because we have three groups for one factor and a continuous outcome, we used a one-way ANOVA comparing mean times across three levels.

The Variables:

1. Independent Variable (IV): Comment Density with 3 levels:

- A (Low Density): Code has very few, basic comments.
- B (Medium Density): Code has clear comments that explain the major sections and complex logic.
- C (High Density): Code is cluttered with too many comments.

2. Dependent Variable (DV): Time to Fix the Bug: We recorded the total time (in minutes) it took for the developer to identify and fix the assigned bug successfully. Lower times mean better code maintainability.

The Hypothesis: Null Hypothesis (H_0): There is no real difference in the average time it takes to fix the bug across the three comment styles.

2. Data (imagined):

Each developer was randomly assigned to one version of the same program (Low/Medium/High comment density). They received the same bug description and success criteria, and a timer started when they opened the file and stopped when tests passed. We saved the fix time (minutes) for analysis.

A Low: 18, 22, 20, 24, 19, 21, 23, 17, 25, 20.

B Medium: 14, 15, 13, 16, 15, 14, 12, 16, 13, 14.

C High: 19, 18, 17, 21, 20, 22, 19, 18, 20, 21.

3. One-way ANOVA calculation steps:

3.1. Group Means & Grand Mean:

A. Low: $\sum X_A = 209$, $\bar{X}_A = 209/10 = 20.90$

B. Medium: $\sum X_B = 142$, $\bar{X}_B = 142/10 = 14.20$

C. High: $\sum X_c = 195$, $\bar{X}_c = 195/10 = 19.50$

Grand Mean = (209 + 142 + 195) / 30 = 18.20

3.2. Total Sum of Squares SS_{Total}:

Formula: $SS_{Total} = \sum (X_i - \bar{X}_G)^2$

A.

Value X_A	Grand Mean	Deviation from Grand Mean	Squared Deviation
18	18.2	-0.2	0.04
22	18.2	3.8	14.44
20	18.2	1.8	3.24
24	18.2	5.8	33.64
19	18.2	0.8	0.64
21	18.2	2.8	7.84
23	18.2	4.8	23.04
17	18.2	-1.2	1.44
25	18.2	6.8	46.24
20	18.2	1.8	3.24

$SS_{A/T}$ = 133.80

B.

Value X_B	Grand Mean	Deviation from	Squared Deviation
14	18.2	-4.2	17.64
15	18.2	-3.2	10.24
13	18.2	-5.2	27.04
16	18.2	-2.2	4.84
15	18.2	-3.2	10.24
14	18.2	-4.2	17.64
12	18.2	-6.2	38.44
16	18.2	-2.2	4.84
13	18.2	-5.2	27.04
14	18.2	-4.2	17.64

$SS_{B/T}$ = 175.60

C.

Value X_C	Grand Mean	Deviation from Grand	Squared Deviation
19	18.2	0.8	0.64
18	18.2	-0.2	0.04
17	18.2	-1.2	1.44
21	18.2	2.8	7.84
20	18.2	1.8	3.24
22	18.2	3.8	14.44
19	18.2	0.8	0.64
18	18.2	-0.2	0.04
20	18.2	1.8	3.24
21	18.2	2.8	7.84

$SS_{C/T} = 39.40$

$$SS_{Total} = SS_{A/T} + SS_{B/T} + SS_{C/T} = 133.80 + 175.60 + 39.40 = 348.80$$

3.3. Within-groups Sum of Squares SS_{group}:

Formula: $SS_{within} = \sum (X_i - \bar{X}_{group})^2$

A.

Value X_A	$\bar{X}_A$	Deviation	Squared Deviation
18	20.9	-2.9	8.41
22	20.9	1.1	1.21
20	20.9	-0.9	0.81
24	20.9	3.1	9.61
19	20.9	-1.9	3.61
21	20.9	0.1	0.01
23	20.9	2.1	4.41
17	20.9	-3.9	15.21
25	20.9	4.1	16.81
20	20.9	-0.9	0.81

$SS_A = 60.90$

B.

Value X_B	$\bar{X}_B$	Deviation	Squared Deviation
14	14.2	-0.2	0.04
15	14.2	0.8	0.64
13	14.2	-1.2	1.44
16	14.2	1.8	3.24
15	14.2	0.8	0.64
14	14.2	-0.2	0.04
12	14.2	-2.2	4.84
16	14.2	1.8	3.24
13	14.2	-1.2	1.44
14	14.2	-0.2	0.04

$SS_B = 15.60$

C.

Value Xc	$\bar{X}_c$	Deviation	Squared Deviation
19	19.5	-0.5	0.25
18	19.5	-1.5	2.25
17	19.5	-2.5	6.25
21	19.5	1.5	2.25
20	19.5	0.5	0.25
22	19.5	2.5	6.25
19	19.5	-0.5	0.25
18	19.5	-1.5	2.25
20	19.5	0.5	0.25
21	19.5	1.5	2.25

$SS_C = 22.50$

$SS_{within} = 60.90 + 15.60 + 22.50 = 99.00$

3.4. Between-group Sum of Squares $SS_{Between}$:

Formula: $SS_{Between} = \sum[(\bar{X}_g - \bar{X}_G)^2 n]$

A. $(\bar{X}_A - \bar{X}_G)^2 n = (20.9 - 18.2)^2 \times 10 = (2.7)^2 \times 10 = 7.29 \times 10 = 72.90$

B. $(\bar{X}_B - \bar{X}_G)^2 n = (14.2 - 18.2)^2 \times 10 = (-4.0)^2 \times 10 = 16.00 \times 10 = 160.00$

C. $(\bar{X}_C - \bar{X}_G)^2 n = (19.5 - 18.2)^2 \times 10 = (1.3)^2 \times 10 = 1.69 \times 10 = 16.90$

$SS_{Between} = 72.90 + 160.00 + 16.90 = 249.80$

3.5. Degrees of Freedom:

k = Number of groups = 3
N = Total number of observations = 30
- Between-Groups Degrees of Freedom: $df_B = k-1 = 3 - 1 = 2$
- Within-Groups Degrees of Freedom: $df_W = N - k = 30 - 3 = 27$
- Total Degrees of Freedom: $df_T = N - 1 = 29$

3.6. Mean Squares (MS):

$$\mathbf{MS_B = SS_{Between} / df_B = 249.80 / 2 = 124.90}$$

$$\mathbf{MS_W = SS_{within} / df_W = 99.00 / 27 = 3.670}$$

3.7. F Statistic:

$$\mathbf{F = MS_B / MS_W = 124.90 / 3.67 = 34.06}$$

Using the F-distribution calculator with F(2,27) = 34.06

$$p = 4.138 \times 10{-}8 = 0.00000004138 \text{ (report as } p<.001)$$

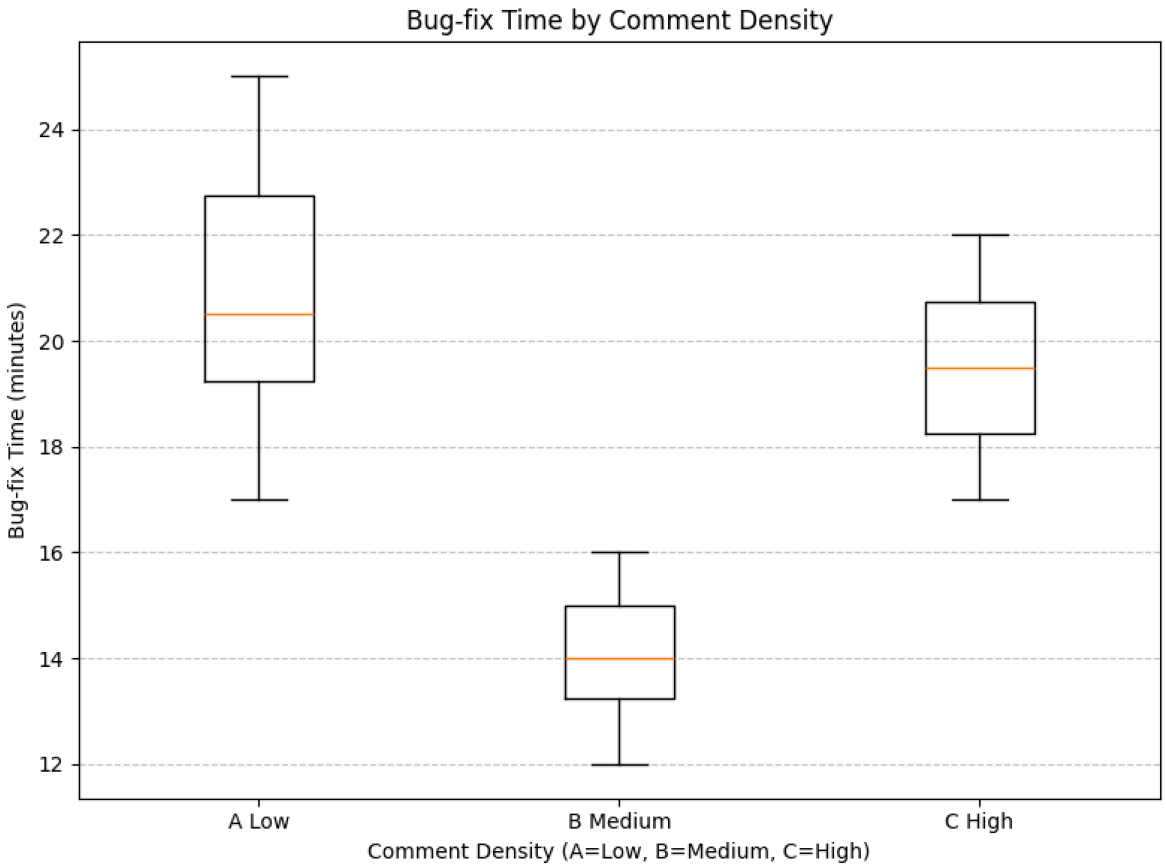
Decision: Reject H_0. That means that comment density had a significant effect on bug-fix time.

4. Report the result:

The results of the one-way ANOVA revealed that the density of inline code comments had a clear impact on the time it took developers to fix a bug. On average, developers working with code containing:
-**low comment density** required about 20.9 minutes, showing moderate variability between participants (Standard Deviation ≈ 2.60).
-**Medium comment density** was considerably **faster**, completing the task in an average of 14.2 minutes with relatively low variability (Standard Deviation ≈ 1.32).
-**High comment density** group performed somewhat better than the low-density group but still slower than those in the medium condition, with a mean bug-fix time of 19.5 minutes and variability of about 1.58 minutes.

==> Just by looking at the averages, Medium comments seem best for speed.
==> The effect of comment density on fixing bugs is statistically significant.
==> Developers worked fastest when comments were clear and concise (Medium density).
==> Too few comments (Low density) or too many comments (High density) both slowed them down.
==> This suggests that balance is key: helpful comments speed up understanding, but clutter makes code harder to follow.

5. Box-plot graph:



I used **Python (matplotlib)** to generate my boxplot comparing bug-fix times across three levels of comment density (Low, Medium, High). The visualization shows that the **Medium comment density group (B)** has the shortest and most consistent bug-fix times, indicated by a lower median and smaller spread. In contrast, the Low (A) and High (C) groups have higher medians and wider ranges, suggesting more variability in fix times. This indicates that a moderate level of comments provides the best balance, making bug-fixing faster and more predictable.